

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO4: Implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS (BL3)
Assessment Tool Weightage: 30%

QUESTIONS: 15

Total Marks:30

Annexure A

CLASS TEST

Code of Conduct:

*I **N. Murali sai (192372309)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Murali sai. N

QUESTIONS:**Total Marks: 30**

Q1. Define architectural patterns in object-oriented systems. List any two common patterns. Explain how the Model-View-Controller (MVC) architecture improves modularity and maintainability.

[L1, L2, L3]

Q2. What is the Data Access Object (DAO) Pattern? Mention two advantages. Explain how DAO separates database-related operations from application logic.

[L2, L3, L4]

Q6. Explain the structure of a three-tier architecture consisting of Presentation, Business, and Data layers. Analyze the flow of a customer transaction through these layers in an e-commerce application.

[L2, L4, L5]

Q9. What is encapsulation in an object-oriented system? Explain how encapsulation protects object state and compare it with data hiding using a suitable example.

[L2, L4]

Q15. Design a layered architecture for a Library Management System. Identify suitable layers, recommend relevant design patterns, and justify how your proposed architecture supports flexibility and maintainability.

[L3, L4, L5]

20/8/26

OOAD (CSA1110)

CO4 - AT 1

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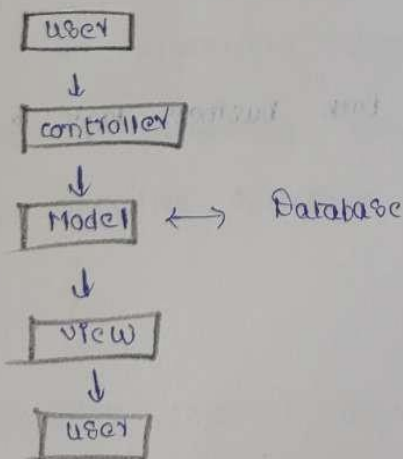
1. Architectural patterns and MVC

- * An architectural pattern is a reusable, high level solution for organising the major components of a software system & defining how those components communicate with each other.
- * It provides a standard structure for designing software systems
- * Two common Architectural patterns
 - Model view controller
 - Three tier Architecture

Model view controller (MVC)

- * MVC divides a application into three major components
 - 1) Model - Represent data / business or domain information.
 - 2) View - Responsible for displaying information to the user.
 - 3) controller - Handles user requests and coordinates b/w the model and view

In an online shopping application,



- * Developers can modify the view without changing the entire business logic.
- * MVC improves modularity, maintainability & separation of concerns by dividing a application into independent but cooperative components.

2) Data Access object (DAO):-

The Data access object pattern is a design pattern used to separate database access operations from the application's business logic.

* A DAO provides methods ^{for} operation such as

Insert search
Delete update

Basic structure:

Application logic

↓

DAO

↓

Database

* For example:

The application does not need to know how SQL queries are written.

Advantages: Separability of concerns and Maintainability

DAO separates Database operations:

consider an online shopping system

without DAO:

Order service

↓

SQL queries

↓

Database

* The order service contains both business logic & database code

with DAO:

Order service

↓

Order DAO

↓

Database

* The order services focuses on order processing, while orderDAO handles database operations

3. Encapsulation and Data Hiding:

Encapsulation: It is process of combining data and the methods that operate on that data inside a class while controlling access to the object's internal state.

Example:

```
class BankAccount {  
    private double balance;  
    public void deposit (double amount) {  
        if (amount > 0) {  
            balance = balance + amount;  
        }  
    }  
}
```

* private double balance: prevents direct access from outside the class

* without encapsulation: account balance = -5000;
could potentially create a invalid state

* with encapsulation: The class controls how the balance changes.

- Data hiding: combines data and behaviour in a class
Restricts access to internal details

* Focuses on organizing responsibilities

* Focuses on preventing unwanted access achieved through classes controlled interface achieve using private, protected.

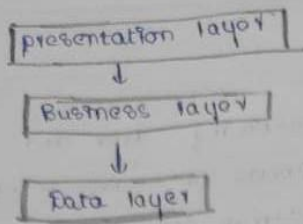
Conclusion:

* Encapsulation protects object integrity

* Data hiding prevents direct access to implementation details.

4) Three tier architecture in a Ecommerce application:-

A Three tier architecture divides an application into three major layers



1) Presentation layer:

This is the user interface

Example: web page

Mobile application

Product screen

Responsibilities:

Accept user input, Displays result, send req to business layer

2) Business layer:

This contains application and business processing

Example: order processing

price calculation

payment processing

Discount

Inventory checking

3) Data layer:

This communicates with persistent storage

Examples:

product database

customer database

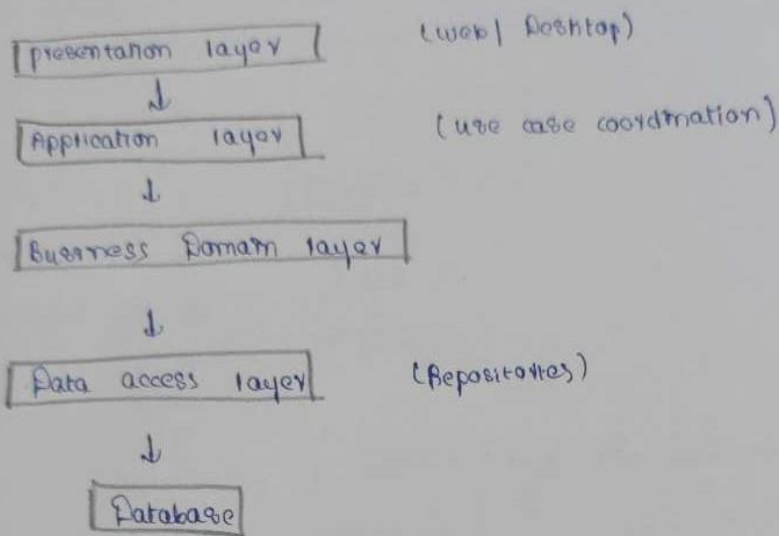
order "

* This architecture creates a clear separation b/w 3 layer & making application easier to develop & maintain

6.

Library Management System:

Architecture Diagram From LMS



1) Presentation layer

Responsible for login screen
Book search

2) Application layer:

Coordinate use case such as
Issue book
Return / entry book

3) Business layer:

If book is not available don't issue the book

4) Database:

Manager the data of user

Books
login details

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand